BUDI95 Fuel Subsidy Policy: Real-Time Sentiment Analysis on Malaysian YouTube Comments

Presented by Fast & Furious

Introduction

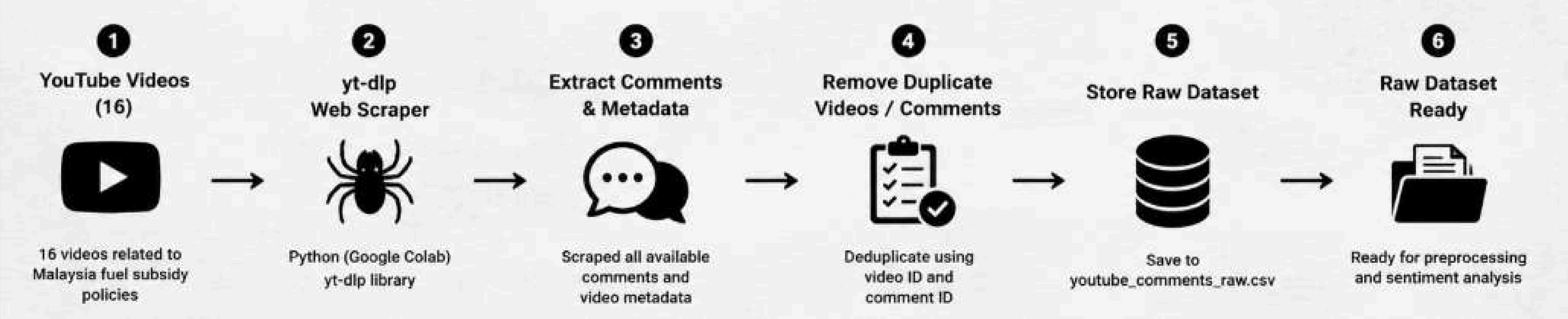
- BUDI95 policy sparked widespread public discussion on YouTube
- High volume of comments makes manual analysis inefficient
- Comments are multilingual — English, Malay, and Manglish (code-switched)
- Solution: automated real-time sentiment classification pipeline

Objectives

- Collect and clean YouTube comments for sentiment analysis
- Develop and evaluate sentiment classification models using machine learning and transformer-based approaches
- Compare model performance using Accuracy, Precision, Recall and F1-Score
- Implement a real-time sentiment analysis pipeline with Apache Kafka and Spark Structured Streaming
- Compare batch processing vs streaming processing performance

Data Collection

- Comments were collected from 16 YouTube videos related to Malaysia's fuel subsidy policies using yt-dlp. The scraper extracted all available comments and metadata, removed duplicate entries, and stored the data as a raw dataset for preprocessing.



Data Collection

📊 DATASET SUMMARY

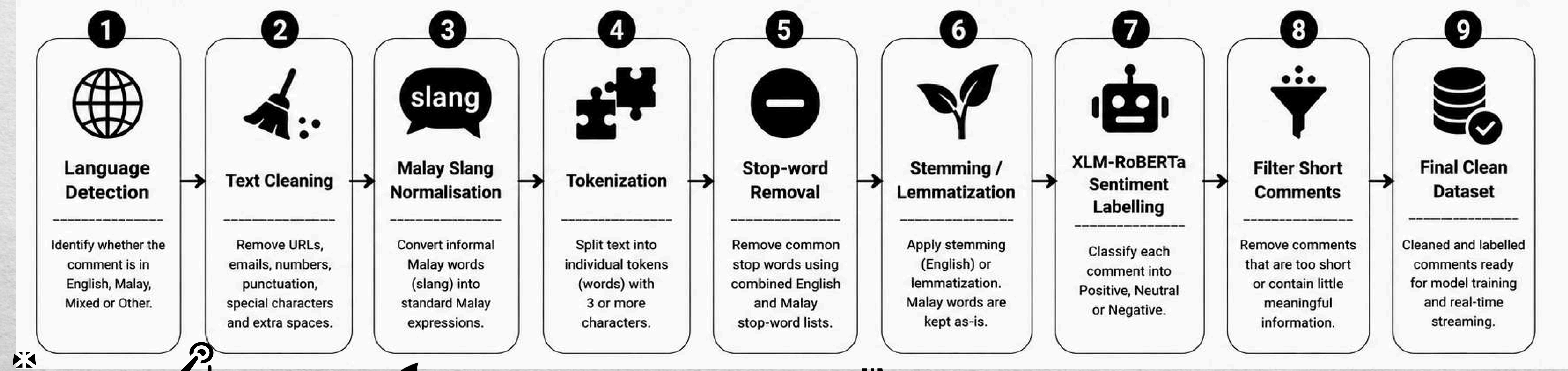
```
Total comments (raw)       : 3,448
Duplicate comments removed: 0
Final unique comments      : 3,448
```

```
 comment_type breakdown:
comment_type
top_level     1778
reply         1670
```

	comment_id	parent_id	comment_type	author	text	like_count	reply_count	published_at	video_id	video_title
0	UgwD0xAhOkfF5igvEQF4AaABAg	None	top_level	@Hasif_Nidzam	Say NO! to subsidy for destruction. GO! green ...	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as bla...
1	Ugw66ZF14ReAbgVlgbx4AaABAg	None	top_level	@asaizanm	in malaysia . the reality is far beyong nightm...	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as bla...
2	UgzmliDCx57h-93MWGB4AaABAg	None	top_level	@ngrobert5054	Since the lowest you pay for us sohai	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as bla...
3	UgxCaZX3Zc_dUVAH0sh4AaABAg	None	top_level	@yeongvoonkang1966	Diesel price didn't ' jumped', it was 'normali...	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as bla...
4	UgyzxxHFCcnzvFEFVql4AaABAg	None	top_level	@maxbuzz4061	alot of business will seize this golden opport...	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as bla...

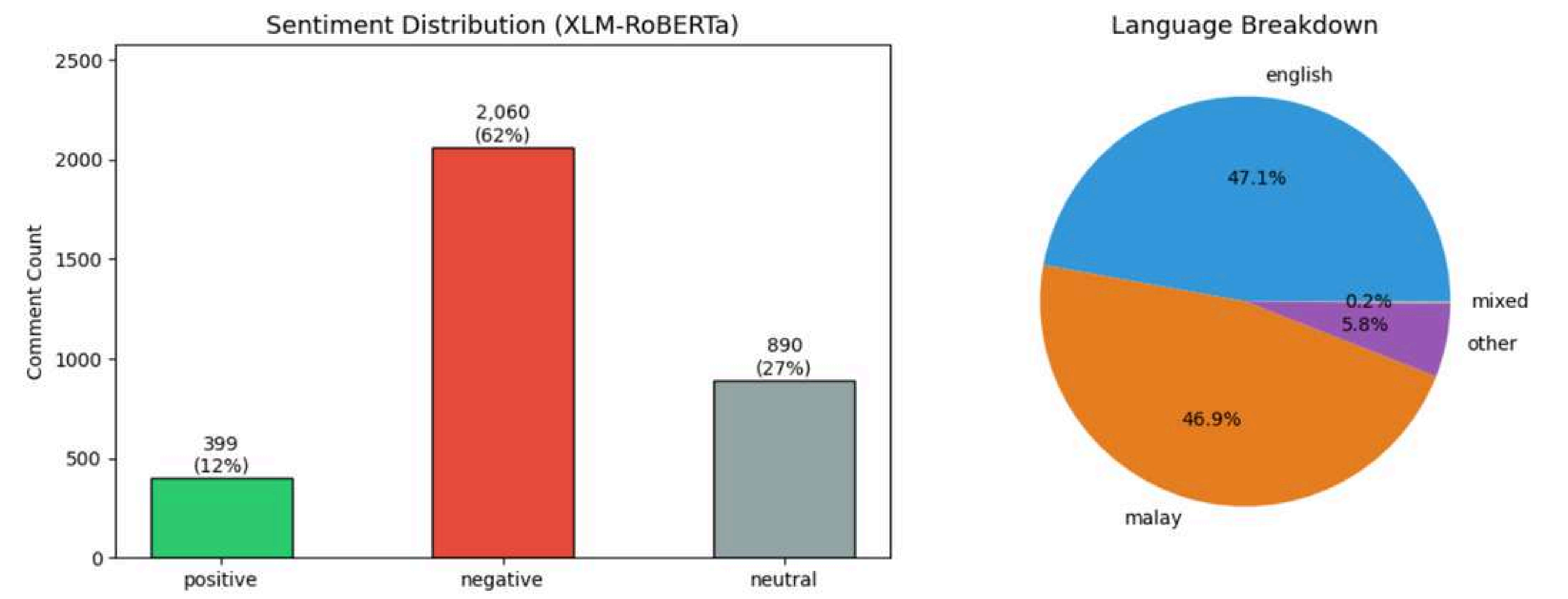
Data Preprocessing

- Raw comments collected from YouTube are cleaned and transformed through a series of natural language processing steps before sentiment labelling using XLM-RoBERTa.



Cleaned Data

```
Total comments (final)      : 3,349
Positive                    : 399 (11.9%)
Negative                    : 2,060 (61.5%)
Neutral                     : 890 (26.6%)
```



comment_id	comment_type	author	language	cleaned_text	sentiment	polarity	prob_neg	prob_neu	prob_pos	like_count	reply_count	published_at	video_id	video_title
5igvEQF4AaABAg	top_level	@Hasif_Nidzam	english	say subsidi destruct green technolog hail natu...	positive	0.579366	0.130316	0.160001	0.709683	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as bla...
AbgVlgbx4AaABAg	top_level	@asaizanm	english	malaysia realiti far beyong nightmar that	negative	-0.420676	0.543132	0.334412	0.122456	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as bla...
3MWGB4AaABAg	top_level	@ngrobert5054	english	sinc low pay sohai	neutral	-0.242629	0.358122	0.526385	0.115493	0	0	2025-06-13T00:00:00+00:00	LmfFs0KWJjA	Diesel prices jump over 50% in Malaysia as

Model Development

- **Naive Bayes** — TF-IDF + Laplace smoothing (α=0.1)
- **Logistic Regression** — TF-IDF + balanced class weighting (C=0.5)
- **XLM-RoBERTa** — Multilingual transformer, fine-tuned with weighted cross-entropy loss + early stopping

Model Comparison

Model	Accuracy	Weighted F1
Naive Bayes	0.6388	0.5987
Logistic Regression	0.6373	0.6357
XLM-RoBERTa ⭐	0.7403	0.7381

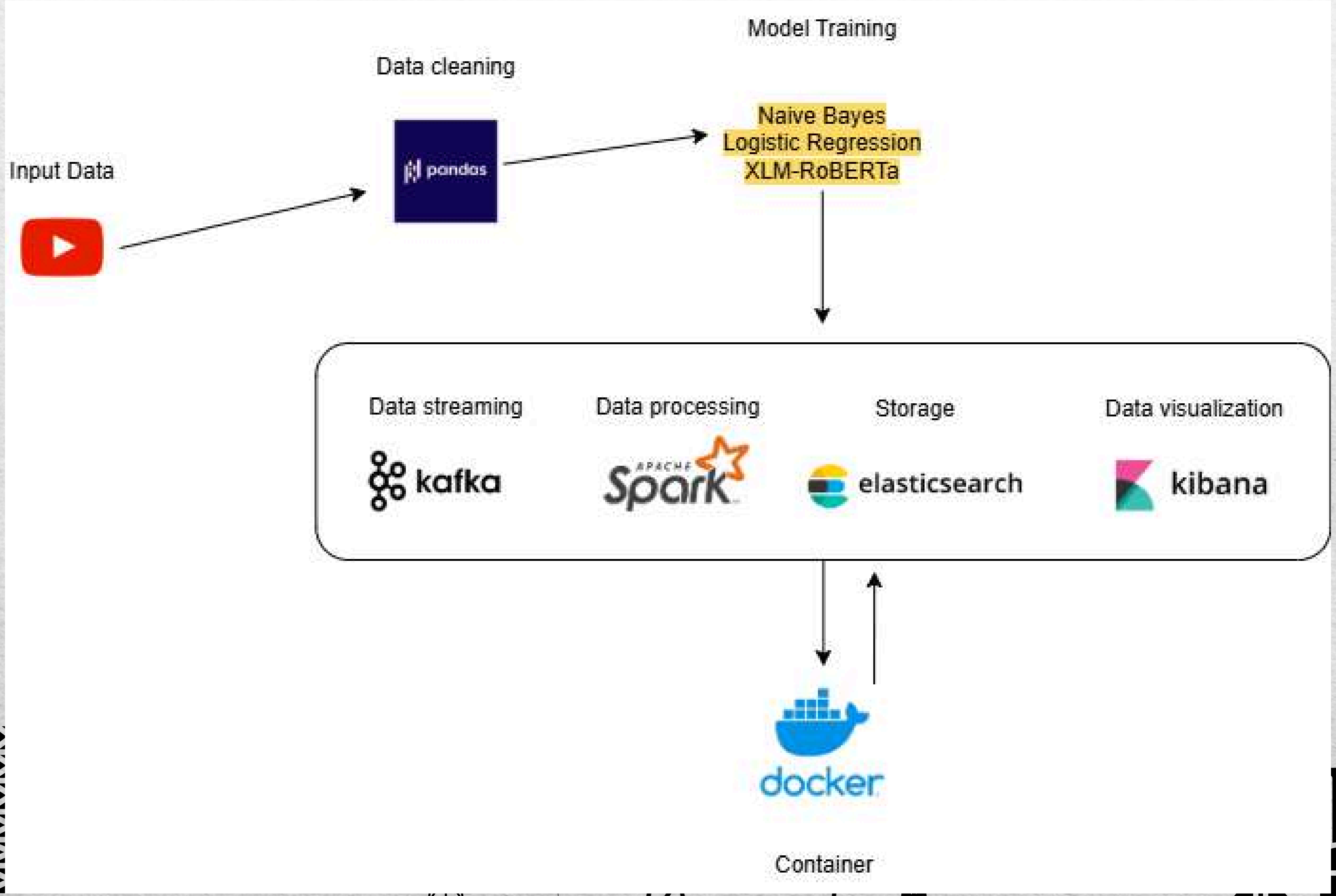
BEST MODEL

Transformers Docs

XLM-RoBERTa

huggingface.co/docs/transformers

Apache System Architecture



1. **Apache Kafka:** It decouples the data source form processing engine, receiving incoming YouTube comments sequentially and hosting them in a distributed queue topic.

2. **Apache Spark:** Used Spark Structured Streaming to ingest micro-batches of data from Kafka in real time.

1. **Elasticsearch:** Serves as the distributed search and analytics engine.

Analysis Result

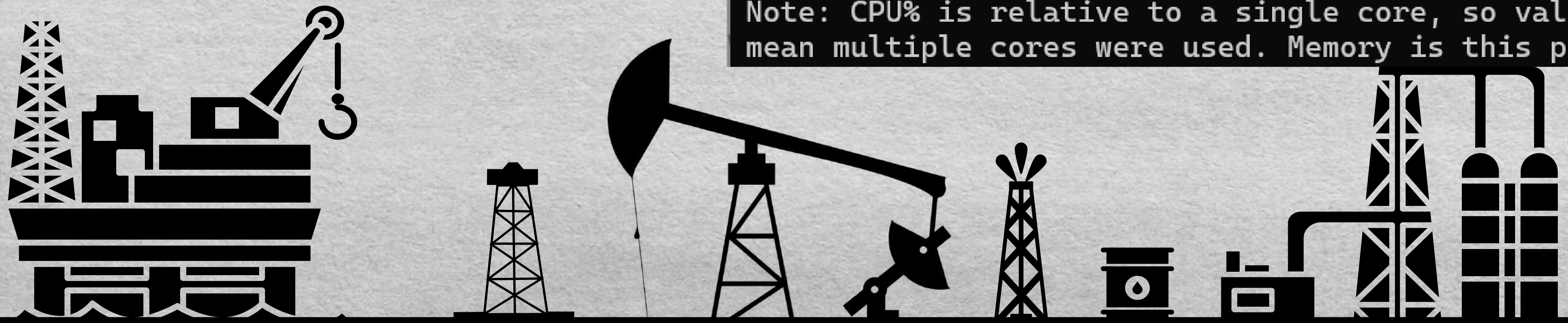
Streaming vs Batch Analysis

Stream Mode

```
  warnings.warn(
[sentiment_model] loaded 'sklearn' (classical) from saved_models
[batch 1]  200 records |  3.798s |     52.7 rec/s -> ES
[batch 2] 3149 records |  0.175s |  17998.9 rec/s -> ES
```

Batch Mode

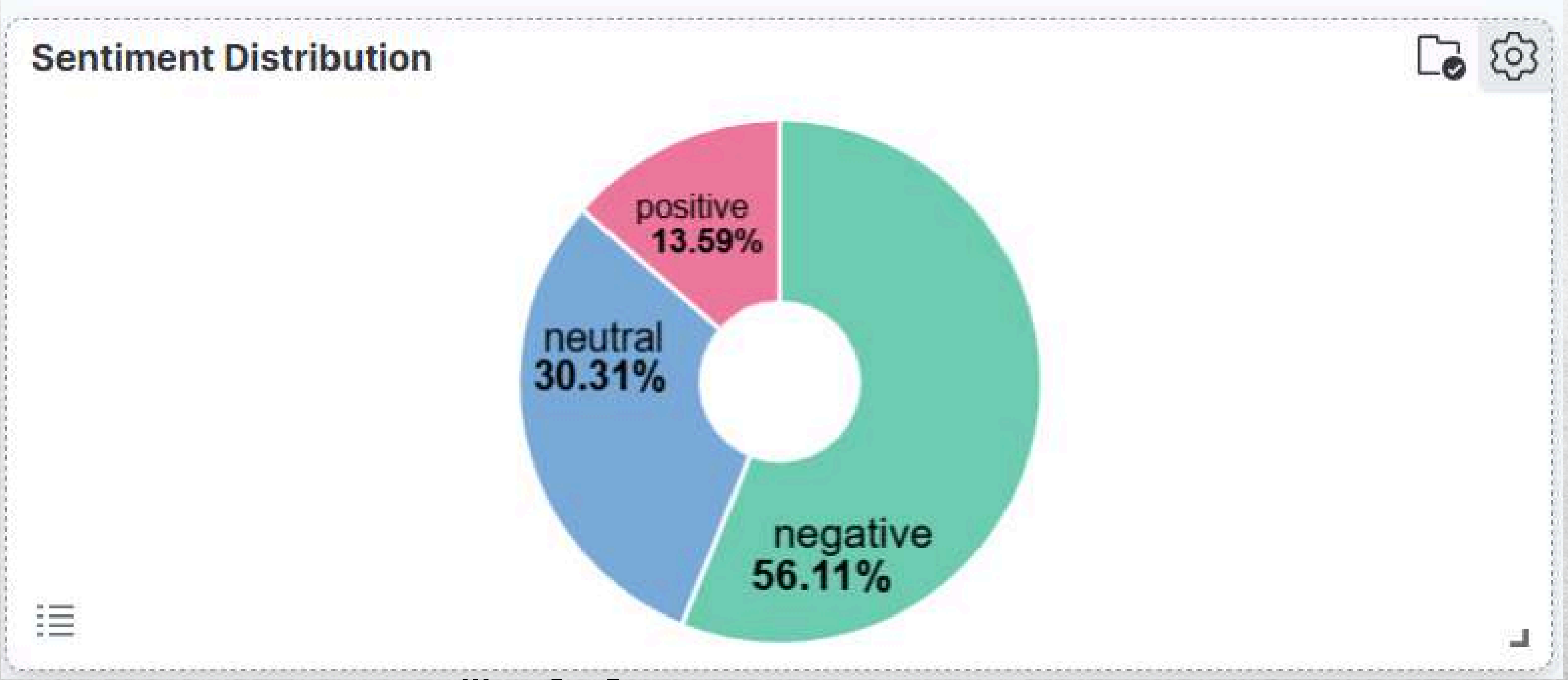
```
==================================================
  BATCH MODE — PERFORMANCE + RESOURCES
==================================================
  Records processed : 3,349
  Total time        : 0.065 s
  Throughput        : 51506.2 records/sec
  Accuracy          : 0.8119
  CPU usage         : 86.9%  (of one core)
  Peak memory (RSS) : 144.0 MB
==================================================
Note: CPU% is relative to a single core, so values above 100%
mean multiple cores were used. Memory is this process only.
```



Analysis Result

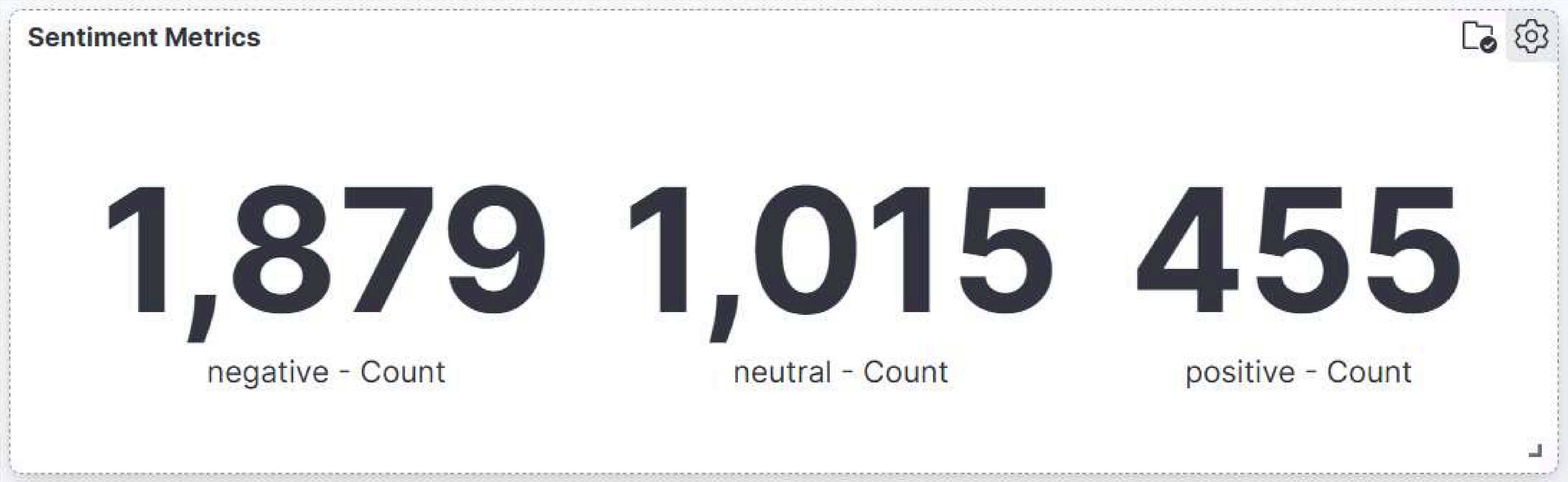
Key Insights

Insight 1: Public sentiment toward BUDI95 is predominantly negative



Analysis Result

Key Insights

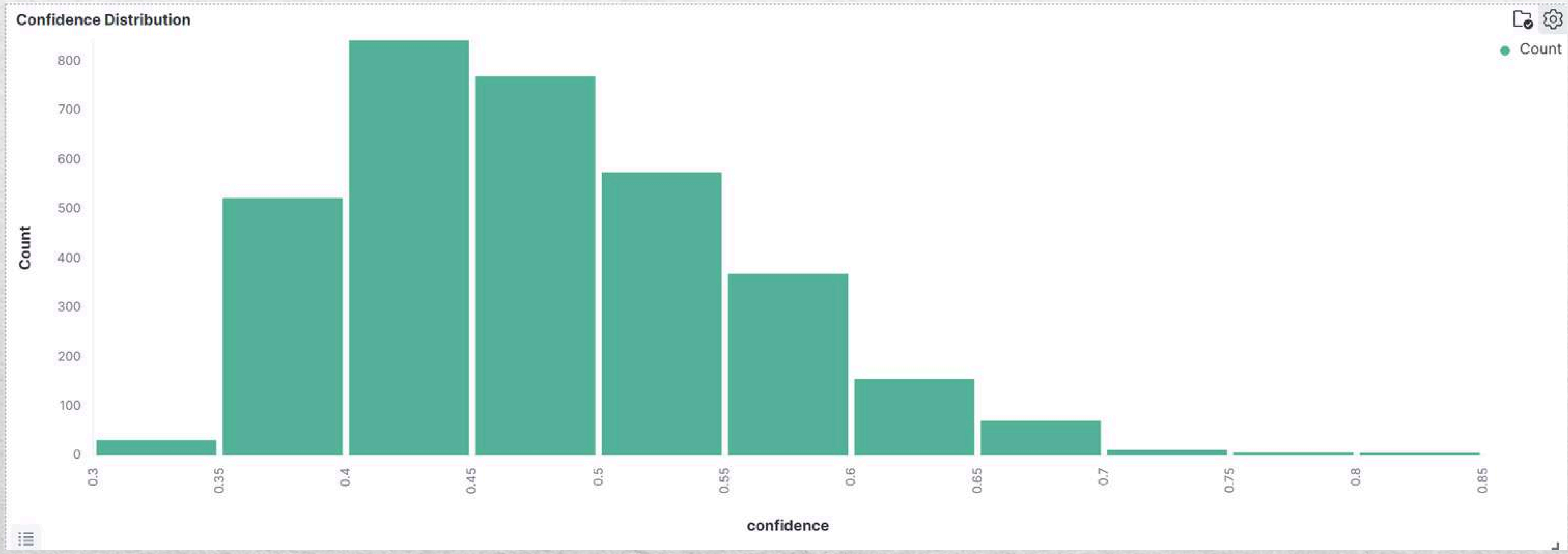


Insight 2: The volume and scale of sentiment is confirmed by absolute counts

Analysis Result

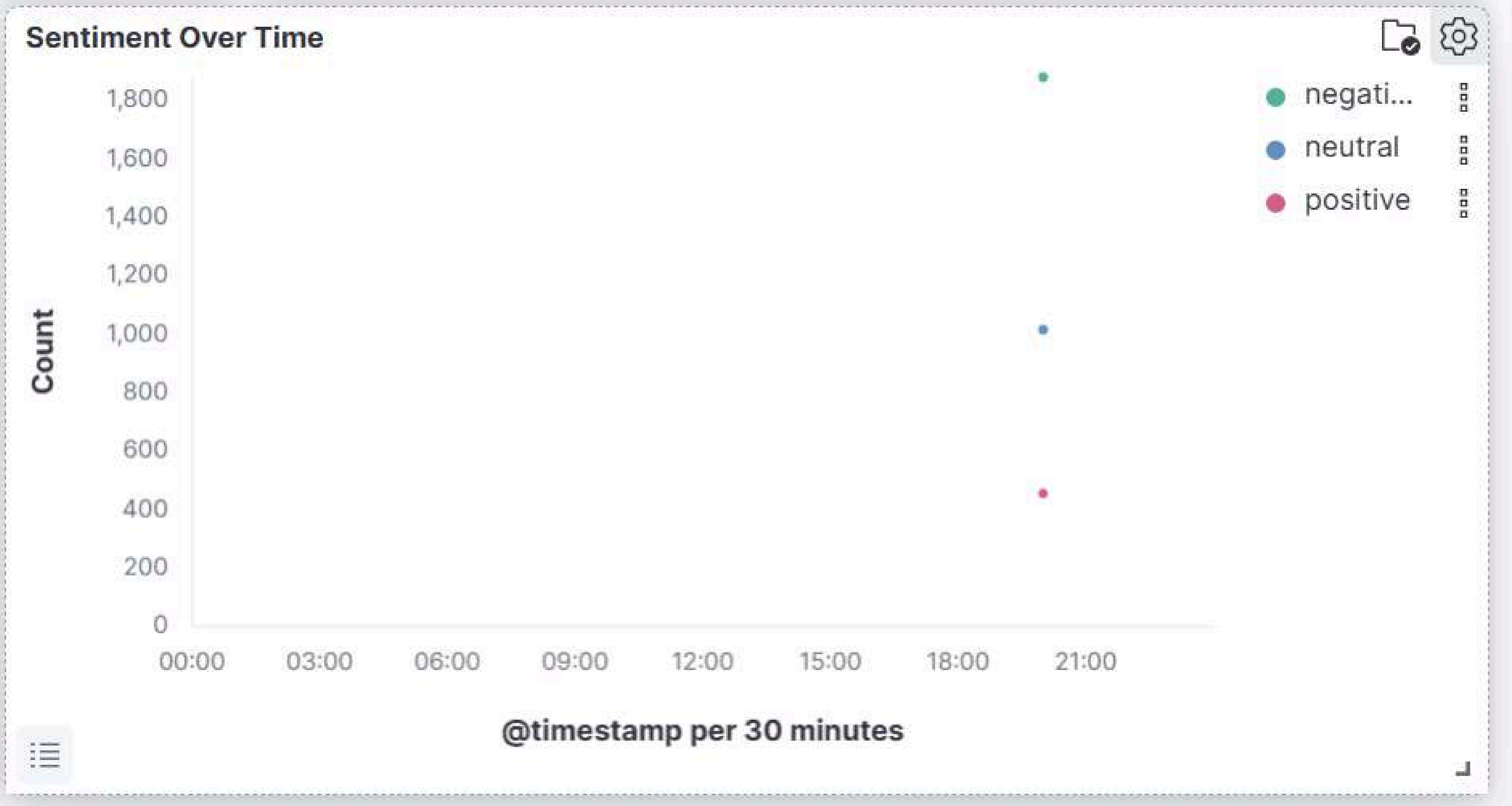
Key Insights

Insight 3: Model confidence is moderate and reflects a balanced three class classifier



Analysis Result

Key Insights



Insight 4: Sentiment data was ingested successfully in real time

Conclusion

- Collected and preprocessed 3,349 multilingual YouTube comments for sentiment analysis.
- XLM-RoBERTa achieved the best classification performance with 74.03% Accuracy and 0.7381 Weighted F1-score.
- Logistic Regression enabled efficient real-time sentiment inference within the streaming pipeline.
- Successfully implemented an end-to-end pipeline using Apache Kafka, Apache Spark Structured Streaming, Elasticsearch, and Kibana with a classification accuracy of 81.19% in both batch and streaming modes.
- Kibana dashboards revealed that 56.11% of comments expressed negative sentiment towards the BUDI95 fuel subsidy policy.

Future Work

Expand Data Sources

- Include Facebook, X (Twitter), TikTok, Reddit and online news comments.
- Increase dataset diversity and coverage.

Improve AI Model

- Fine-tune XLM-RoBERTa using a larger Malaysian social media corpus.
- Explore XLM-RoBERTa-Large, mBERT, or other multilingual transformer models.

Real-Time Data Ingestion

- Integrate the YouTube Data API v3 with Kafka to enable live streaming of newly posted comments.
- Support continuous real-time monitoring of public sentiment during policy announcements.

High-Performance Deployment

- Deploy the streaming pipeline on GPU-enabled cloud platforms such as AWS EMR, Google Cloud Dataproc, or Databricks.
- Enable XLM-RoBERTa inference directly within the streaming pipeline.

Thank You